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SUPERSEDING
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2 July 1974

MILITARY SPECIFICATION

TARGET SET, SURVEYING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a target set used in surveying.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-L-111/38	- Lamp, Incandescent, Miniature, Trade No. 14, G-3 1/2 Clear Bulb, Miniature Screw Base, C-2R Filament.
QQ-B-613	- Brass, Leaded and Nonleaded; Flat Products. (Plate, Bar, Sheet, and Strip).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

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QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-B-18/9	- Battery, Dry, BA-30.
MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.
MIL-T-14091	- Tripod, Surveying: Extension-Leg-Type.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

MI

1010-1814	- Eyepiece Adapter.
1010-2202	- Eyepiece Adapter.
1039-4848	- Hood Light Concealing.

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM B16	- Free-cutting Brass Rod, Bar, and Shapes for use in Screw Machines.
ANSI/ASTM B85	- Aluminum-Alloy Die Castings.
ANSI/ASTM B151	- Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Rod and Bar.
ANSI/ASTM B584	- Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

HS1086	- Unified Numbering System for Metals and Alloys.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The target set shall consist of the following components:

Item	Quantity
Battery box	1
Cable, electrical	1

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Item	Quantity (Continued)
Case, carrying	1
Illumination assembly	1
Lamp, hand	1
Target assembly	1
Tribrach	1

3.1.1 Optional accessories. When specified (see 6.2), one or more of the following components shall be included with the target set:

Item	Quantity
Eyepiece adapter assembly	1
Hood, light-concealing	1
Tripod, surveying	1

3.1.2 Interfacing. The make, model and type of theodolite with which the target set is to be used shall be as specified (see 6.2). Targets and theodolites shall be interchangeable in a common tribrach assembly.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the target set. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, materials and material finishes, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), the contractor shall furnish two target sets for first article inspection and approval (see 4.4 and 6.3).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government two target sets for inspection as specified in 4.5.

3.4 Material. Material shall be as specified herein and on the drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.8).

3.4.1 Material deterioration and control. The target set shall be fabricated from compatible materials, inherently corrosion resistant or suitably treated to

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provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environment to which the item may be exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize galvanic corrosion. Insulation shall be provided by a barrier such as a coat of epoxy primer conforming to MIL-P-23377 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion shall also be obtained by exclusion of the electrolyte if feasible.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or his designated representative.

3.4.2 Unified Numbering System (UNS). Where applicable, metal alloys are identified by UNS in accordance with SAE H51086.

3.5 Threads. All screw threads shall conform to FED-STD-H28.

3.6 Fastening screws, bolts, washers and nuts. Size, type, form and class of fit for fastening screws shall be selected by the contractor as best suited for their intended use. There shall be no exposed threads on the target set except tapped holes.

3.7 Target. The target consists of the following subcomponents: a target plate carrier which clamps to a tribrach and includes a target level, a target plate with a centering mark, a screw-in lamp assembly and a reflector. Tolerances for given dimensions shall be plus or minus 0.005 inch (0.13 millimeter (mm)).

3.7.1 Target plate. The target plate shall consist of the frame, the target mask, and the cover plate. A quick-release mechanism shall be incorporated into the plate for quick removal of the target plate from the target plate carrier.

3.7.1.1 Target frame. The target frame shall be corrosion-resistant metal (CRM), finished in flat black enamel. Overall dimensions for the target frame shall be not more than 4.500 inches (115 mm) in height or width and 0.350 inch (9 mm) in depth. A centering mark shall be located at the exact center of the top rim of the target frame and shall be finished in white gloss enamel.

3.7.1.2 Target mask. The target mask shall be the same CRM as the frame and shall have the same flat black enamel finish. Thickness of the plate shall be not more than 0.04 inch (1 mm). The mask shall fit snugly inside the target frame without binding. The mask shall have three triangular cut-outs arranged to pro-

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vide a sighting pattern for both horizontal and vertical angular measurements when viewed through a theodolite. The triangular cut-outs and their positions on the mask may be any acceptable in general commercial practice.

3.7.1.3 Cover plate. The cover plate shall be frosted translucent glass or plexiglass. The thickness of the cover plate shall be not more than 0.08 inch (2 mm). The cover plate shall fit snugly inside the target frame and on top of the target mask. The cover plate, mask, and frame shall be fastened together by six screws.

3.7.2 Target plate carrier. The target plate carrier shall be CRM and its exterior surfaces finished as specified in 3.16. The target carrier shall be not more than 4.33 inches (110 mm) in total height. The target carrier shall be designed to accommodate the target plate at the top and the tribrach at the bottom. The midsection shall provide a level housing and a bearing/race assembly to permit the target plate and level housing to be rotated in azimuth 360 degrees in both clockwise and counter-clockwise directions.

3.7.3 Target level assembly. The level housing portion of the target plate carrier shall be fitted with a target level consisting of a liquid-filled tubular level vial, a protective clear plastic cover, and an adjusting mechanism.

3.7.3.1 Level vial. A glass level vial with brass end caps shall be provided. The level vial and end caps shall fit securely within the level vial housing. A protective, clear plastic cover shall be fitted over the level vial without binding or giving tension to the glass of the vial. The level vial shall be graduated at 2 mm intervals and shall have a sensitivity of 60 arc-seconds per 2 mm division, plus or minus 15 arc-seconds per division. Graduations shall be symmetrical about the exposed portion of the vial and shall extend far enough to include both ends of the bubble at all operating temperatures specified herein. Graduations shall be certified by the manufacturer to be fired ceramic or filled with indelible pigment, at the manufacturer's option.

3.7.3.2 Adjusting mechanism. The bracket and arm of the level vial housing shall be fitted with a spring-loaded capstan adjusting screw.

3.7.4 Tribrach assembly. A tribrach assembly shall be provided and shall have the following major components: base plate, plate spring, lock plate, leveling screws, optical plumbing device, tribrach, and circular level. All major components shall be diecast corrosion-resistant alloys conforming to ANSI/ASTM B85.

3.7.4.1 Base plate. The three radial arms of the base plate shall be equally spaced at 120 degree intervals and extend 2.875 inches (73.03 mm) from the center point. Three bearing points shall be located on each arm of the base plate 2.375 (60.33 mm) from the center point. The three bearing points shall be case-hardened

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carbon steel and given a black oxide treatment for corrosion resistance. The center of the base plate shall have a 5/8-11 UNC-2B nut.

3.7.4.2 Plate spring. The plate spring shall attach to the base plate and shall be secured with a nut and a lock screw. The spring plate shall conform to QQ-B-613. The plate spring shall be spring tempered and either plated or painted to resist corrosion.

3.7.4.3 Leveling screws. Material for the leveling screws shall conform to ANSI/ASTM B151, UNS 79200, H01 temper. The three adjusting nuts and three milled knobs shall conform to ANSI/ASTM B16, H02 temper. The milled knob shall be chrome plated not less than 0.19 mil thickness. The leveling screws shall permit leveling of the target when the base plate is not more than 6 degrees out of level.

3.7.4.4 Optical plumbing device. An optical plumbing device shall be provided and shall be affixed to the tribrach. The optical plumb shall be adjustable. The reticle pattern shall be two mutually perpendicular etched lines intersecting at the center of the field of view. The optical system shall provide a clear image and clear reticle pattern throughout the entire operating range which shall be from 3 feet (0.914 meters (m)) to not less than 110 feet (33.5m). The eyepiece and milled collar shall extend beyond the tribrach not more than 0.75 inch (19.05 mm). Clear image is defined as no blurring of the image or reticle lines.

3.7.4.5 Base casting, tribrach. The tribrach shall be diecast corrosion-resistant alloy as specified in 3.7.4 and shall be 4.134 inches (105.0 mm) outside diameter. The casting shall include three integral flanges on the outer rim, spaced 120 degrees apart. Each flange shall be drilled and tapped to accommodate the leveling screws and their dust covers, stop screws, set screws, and adjusting screws. Holes for the leveling screws shall be centered 2.364 inches (60.05 mm) from the center point of the tribrach. Clutch bolt holes shall be centered on the same 120 degree radials 1.25 inches (31.75 mm) from the tribrach center point. The eyepiece of the optical plumb shall be attached to the tribrach at a point centered midway between two of the flanges. One flange shall also house the circular level.

3.7.4.6 Circular level. A circular level shall be located in a metal housing, which shall be part of the casting of one of the tribrach flanges. The circular level shall be adjustable and have a sensitivity of 10 arc-minutes per 2 mm division plus or minus 3 arc-minutes per division. The center of the glass vial shall be etched with a black circle 2 mm in diameter.

3.7.4.7 Lock plate. Material shall be ANSI/ASTM B584, UNS C86400 or C86700. The lock plate shall be plated first with nickel, then with chromium for a total thickness not less than 0.19 mil. The lock plate motion may be controlled by a lever or cam-action device made from the same material and plating and mounted on

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the tribrach. The lock plate motion shall secure the clutch bolts of the target base with an error of centering not to exceed 0.002 inch (0.5 mm).

3.8 Illumination assembly and reflector. The illumination assembly shall consist of an incandescent lamp, cable and reflector.

3.8.1 Illumination assembly. The cable shall be not less than 36 inches (914 mm) nor more than 40 inches (1016.0 mm) in length. The cable shall be double strand insulated wire. The insulation shall be black in color. One end of the cable shall be wired to a 2-prong connecting plug. The plug shall mate with the sockets provided in the battery box. The opposite end of the cable shall be wired to an insulated socket threaded to accept one lamp conforming to W-L-111/38, Trade No. 14. Additional threading shall be provided on the socket where it shall mate to the target reflector.

3.8.2 Reflector. The reflector shall be corrosion-resistant metal. Two slotted holes on the upper flange of the reflector shall be provided and shall accommodate the 2 pivots located on the target frame. The back of the reflector shall be equipped with a hollow cylindrical stub, threaded on the outside, which shall accommodate the lamp and threaded socket. The reflector shall be finished in white enamel on all inside surfaces and flat-black enamel on all exterior surfaces.

3.9 Battery box. A battery box and light accessory case shall be provided. Materials for the case may be corrosion-resistant metal, fiberglass-reinforced plastic laminate, or high impact plastic that will withstand the temperature and shock test conditions specified in section 4 without breaking or damage. The case shall provide space for not less than 4 nor more than 6 batteries. Batteries shall conform to MIL-B-18/9, but shall not be provided with the target. Batteries shall be connected in a series-parallel arrangement to provide an output not to exceed 3 volts. The case shall have 2 power output sockets which can be sealed against environmental conditions by a spring-loaded slide. One power output socket shall be wired to a rheostat. The rheostat knob shall be marked with an OFF position. The second output terminal shall be wired for full output voltage. The case shall have a hinged opening and shall provide enough interior space to hold all batteries as well as the hand light, electrical cable and 4 replacement lamps. Two positive clamping latches shall be provided to secure the case in the closed position. A spring-tension clip shall be firmly affixed to the back of the battery box for attaching to the tripod during operation. Unless otherwise specified (see 6.2), the battery box shall be treated and finished in accordance with MIL-T-704, type A. A carrying strap shall be provided, made from webbing conforming to MIL-W-530, type IIa, class 4, dyed olive drab 7. Two dummy batteries shall be furnished.

3.9.1 Hand light and electrical cable. The hand light and electrical cable shall connect to output terminals on the battery box. The hand light shall have

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an ON/OFF switch and connecting cable 6 feet, plus or minus 1 foot (1.8 m plus or minus 0.3 m) in length. The socket of the hand light shall be threaded to accept a miniature lamp conforming to W-L-111/38, Trade No. 14. The opposite end of the hand light cable shall be fitted with a 2-prong plug. The separate electrical cable shall be not less than 36 inches (0.9 m) nor more than 40 inches (1.02 m) in length and shall be wired to 2-prong plugs on both ends.

3.10 Target set carrying case. A carrying case shall be provided. Materials for the case may be plywood, molded thermoplastic resin, fiberglass-reinforced plastic laminate or other materials capable of withstanding the environmental tests specified in section 4 without damaging the target or accessories. The case shall be provided with a carrying strap or handle. The case shall have a hinged lid with a stop device which prevents the lid from dropping or producing excessive torque on the hinge. Corners and edges shall be reinforced and the lid shall be provided with a lockable metal latch. All metal parts shall be corrosion-resistant metal or metal treated to resist corrosion. The interior of the case shall be sectioned with blocking and padding to provide support and protection to the stored components. Each case shall accommodate a tribrach, reflector, target, battery box, cables, light-concealing hood, and eyepiece adapter. The case shall protect all contents with no damage, leakage of water, or exposure to dust when tested as specified in section 4.

3.11 Light concealing hood. When required (see 3.1.1), a light concealing hood shall be furnished. The light concealing hood shall be as specified in MIRADCOM Drawing 1039-4848.

3.12 Eyepiece adapter assembly. When required (see 3.1.1), an eyepiece adapter assembly shall be furnished. The eyepiece adapter assembly shall be as specified in MIRADCOM Drawings 1010-1814 and 1010-2202.

3.13 Tripod assembly. When required (see 3.1.1), a tripod shall be furnished conforming to MIL-T-14091.

3.14 Environmental conditions. The target set shall remain operational under the following environmental conditions.

3.14.1 Operating temperature. The target, tripod, carrying case, and all accessories including wiring shall operate as specified herein and shall not be damaged when subjected to ambient temperatures from plus 110° F (plus 43° C) for 5 hours to minus 25° F (minus 32° C) for 6 hours.

3.14.2 Storage temperature. The target carrying case and accessories shall not be damaged by storage under the following air temperature conditions:

High - plus 145° F (plus 63° C) plus or minus 2° F (1° C). Time period - 1 hour daily.

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Low - minus 28° F (minus 33° C) plus or minus 2° F (1° C). Time period - 6 hours. No solar radiation.

3.14.3 Solar radiation. The target and accessories shall operate as specified herein and shall not be damaged by full impact of solar radiation of 355 British thermal units per square foot per hour at an air temperature of plus 110° F (43° C) plus or minus 2° F (1° C) for 4 hours.

3.14.4 Rainfall. The target and its carrying case shall show no signs of water penetration when subjected to rainfall at a rate of 3 inches (76.20 mm), plus or minus 1 inch (25.40 mm), per hour for a period of 1 hour.

3.14.5 Sand and dust. The target and its carrying case shall not be damaged or show sand and dust penetration in the optics or bearing surfaces of the target when subjected to blowing sand and dirt tests specified in 4.7.2.6.

3.14.6 Vibration and shock. The target and accessories in the carrying case shall not be damaged when tested as specified in 4.7.2.9 and 4.7.2.10. When a tripod is required (see 3.13), it shall be tested for shock and stability as specified in MIL-T-14091.

3.15 Wiring. Cables used to connect the hand light and illumination assembly to the battery box shall be extra flexible and shall show no evidence of damage nor crack at minus 30° F (minus 34° C).

3.16 Treatment and painting. The surface of the target case and accessories normally painted shall first be given an anodized treatment and a chromate chemical conversion treatment, then painted in accordance with MIL-T-704, type A, color as specified (see 6.2). Nonpermissible galvanic couples (dissimilar metals) as defined by MIL-T-704 shall be insulated as specified in MIL-T-704.

3.17 Identification marking. Each target set shall be identified in accordance with MIL-STD-130. The marking of the target set shall be placed on the side of the target plate carrier. Markings shall include: letters "US", contractor's name or trademark, serial number of the target, and year of manufacture. The carrying case shall be marked as above, but shall also include the national stock number assigned to the target set and the contract number under which the target set was procured.

3.18 Lubrication. Parts requiring lubrication shall be lubricated prior to assembly with grease conforming to MIL-G-23827.

3.19 Government-loaned property. When specified (see 6.2), the property specified in 3.1.1 will be loaned by the Government (see 6.4).

3.20 Tolerances. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed herein to insure correct fit, assembly, and operation of the target sets.

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3.21 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.22 Safety. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required and they are not detrimental to safety. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.23 Workmanship. All threads requiring movement, all sliding or rotating parts shall move without binding, backlash or end play. Clamping actions shall be positive, and when unclamped shall not hinder motion. Focusing components shall retain their focus when finger pressure is released. Lenses shall be free from scratches, chips, cloudiness, or air bubbles between cemented elements, and dirt between the glass and film coating. Optics shall show sharpness of focus, distinct imaging, and definition. Liquid in the level vials shall be clear and the vials shall fit in their housings without looseness. Components and assemblies shall show no evidence of forced attachment. Bearing surfaces shall have no scratches or surface blemishes.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable (see table I).

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Table I. Inspection.

Examination or test	Examination or Test Paragraph	Requirements Paragraph
Level Sensitivity	4.7.2.2.1	3.7.3.1
Tribrach interchangeability	4.7.2.2.2	3.1.2
Optical Plumb	4.7.2.3	3.7.4.4
Tribrach	4.7.2.4	3.7.4
Rainfall	4.7.2.5	3.14.4
Sand and dust	4.7.2.6	3.14.5
Operational	4.7.2.7	3.14
Low temperature operation	4.7.2.8.2	3.14.1
Low temperature storage	4.7.2.8.3	3.14.2
High temperature operation	4.7.2.8.4	3.14.1
High temperature storage	4.7.2.8.5	3.14.2
Solar radiation	4.7.2.8.6	3.14.3
Wiring	4.7.2.8.7	3.15
Vibration	4.7.2.9	3.14.6
Shock	4.7.2.10	3.14.6

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Prior-to-assembly inspection (see 4.3).
b. First article inspection (see 4.4).
c. Initial production inspection (see 4.5).
d. Quality conformance inspection (see 4.6).
e. Inspection comparison (see 4.8).
f. Inspection of packaging (see 4.9).

4.3 Prior-to-assembly inspection.

4.3.1 Tests. Prior to assembly, each level vial shall be tested for sensitivity, as specified in 4.7.2.2.1. The focusing test specified in 4.7.2.3.1 also may be conducted prior to assembly.

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4.4 First article inspection.

4.4.1 Examination. The first article target set, carrying case, battery box, and accessories shall be examined as specified in 4.7.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The first article target set, carrying case, battery box, and accessories shall be tested as specified in 4.7.2. Failure of any test shall be cause for rejection.

4.5 Initial production inspection. When specified (see 3.3), two initial production target sets will be selected at random by the Government from the target sets being produced by production tooling and will be examined as specified in 4.7.1 and tested as specified in 4.7.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production target set shall not exclude the remaining target sets from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.5.1 Inspection failure. Failure of an initial production target set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.5 shall be cause for rejection of the initial production target set and shall be cause for refusal by the Government to continue acceptance of production target sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on target sets previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all target sets accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Quality conformance inspection.

4.6.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2 Examination. Samples selected in accordance with 4.6.1 shall be examined as specified in 4.7.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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4.6.3 Tests.

4.6.3.1 Individual. Each target set shall be operated as specified in 4.7.2.7. Each level vial shall be tested as specified in 4.7.2.2.1. Failure of any test shall be cause for rejection.

4.7 Inspection procedure.

4.7.1 Examination. The target set shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Materials not corrosion resistant or suitably treated.
103. Dissimilar metals not in accordance with MIL-STD-889.
104. Material and material finish or treatment information not available.
105. Target set components not as specified.
106. Target frame not as specified.
107. Target mask not as specified.
108. Cover plate not as specified.
109. Level vial graduations not as specified.
110. Level adjusting mechanism not as specified.
111. Target does not rotate as specified.
112. Target plate carrier not as specified.
113. Tribrach not as specified.
114. Leveling screws not as specified.
115. Optical plumb not as specified.
116. Reticle pattern not as specified.
117. Illumination assembly not as specified.
118. Reflector not as specified.
119. Battery box not as specified.
120. Accessories not as specified (hand light and cable).
121. Carrying case not as specified.
122. Light concealing hood not as specified, when applicable.
123. Eyepiece adapter assembly not as specified, when applicable.
124. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification marking missing, incomplete, or illegible.

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4.7.2 Tests.

4.7.2.1 Test conditions. Prior to beginning the tests, the contractor shall make all required adjustments to level vials, set screws, and other moving parts to operate as specified herein. After contractor adjustment of any adjustable parts readjustment shall not be made, except at the conclusion of the shock, vibration, and temperature tests.

4.7.2.2 Prior to assembly.

4.7.2.2.1 Level sensitivity. The sensitivity of each level vial shall be determined by mounting the level vial on a level trier or by any other method yielding values accurate to within 5 percent. The plane of the graduated vials shall be changed by an amount sufficient to move the ends of the bubble not less than 4 mm as measured by the 2 mm graduations on the vial. Evidence that the target level does not have a sensitivity of 60 arc-seconds, plus or minus 4 seconds, or that the circular level of the tribrach does not have sensitivity of 10 arc-minutes, plus or minus 3 minutes, shall constitute failure of this test.

4.7.2.2.2 Tribrach interchangeability. The target shall be removed from its tribrach assembly and mounted into the tribrach of the designated theodolite type (see 3.1.2). The theodolite shall also be mounted into the target's tribrach assembly. Failure of the theodolite to fit in its container with the target tribrach or of the target to fit into the theodolite tribrach shall constitute failure of this test.

4.7.2.3 Optical plumb.

4.7.2.3.1 Focusing. Focus the optical system of the optical plumb on objects at 3 feet (0.914 m), 64 feet (19.5 m), 90 feet (27.4 m), and 110 feet (33.5 m). Blurring of the image or reticle lines at the operating ranges of focus shall constitute failure of this test.

4.7.2.3.2 Eccentricity. The target shall be centered over a ground marker with a plumb bob. The plumb bob shall then be removed and the centering checked by means of the optical plumb. This test shall be repeated with the optical plumb eyepiece repositioned on each side of the tripod head. Evidence that the line of sight is not coincident with the true plumb line shall constitute failure of this test.

4.7.2.4 Tribrach.

4.7.2.4.1 Base plate nut. The base plate nut shall be tested for conformance to FED-STD-H28 with certified plug gages. Evidence of any base plate nut not mating with the gage shall constitute failure of this test.

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4.7.2.4.2 Levelment. The target set shall be mounted on a tripod or trivet which is tilted 6 degrees, plus 0, minus 1/2 degree, from the horizontal. The target shall then be leveled with the 3 footscrews. Inability of the target to be leveled constitutes failure of this test.

4.7.2.5 Rainfall.

4.7.2.5.1 Target. The target mounted on its tribrach shall be placed in a rain testing facility not less than 36 inches (914.40 mm) cube (36 inches high, 36 inches deep, 36 inches wide) (914.40 mm x 914.40 mm x 914.40 mm). The facility shall be equipped with water spray nozzles to provide a uniform flow of water over the test area at a rate of 3 inches (76.20 mm) plus or minus 1 inch, (25.40 mm) per hour. The direction of rainfall shall be variable up to 45 degrees from the vertical. Water temperature shall be plus 52° F (plus 11° C) to plus 68° F (plus 20° C). The target temperature shall be within plus or minus 5° F (3° C) of the water temperature. The target shall be subjected to this test for 5 minutes. Evidence of water inside the sleeve, optical plumb, or internal bearings shall constitute failure of this test.

4.7.2.5.2 Carrying case. The target set, stored in its carrying case, shall be subjected to the test specified in 4.7.2.5.1 for a period of one hour. Evidence of water in the carrying case, internal bearings, or optical plumb shall constitute failure of this test.

4.7.2.6 Sand and dust (see 6.5).

4.7.2.6.1 Target. The target, mounted on its tribrach, shall be placed in a sand and dust testing chamber not less than 36 inches (914.40 mm) cube. The sand and dust (140 mesh silica flour) shall pass through the chamber at a velocity of 100 to 500 feet per minute (30.48 m to 152.4 m per minute). The density of sand and dust shall be maintained at 0.1 to 0.5 gram per cubic foot within the test area. The internal temperature shall be maintained at plus 77° F (plus 25° C), plus or minus 5° F (3° C). The relative humidity shall not exceed 30 percent. The target shall be subjected to this test for 15 minutes. Evidence of sand or dust inside the sleeve optical plumb or internal bearings, or chipping or cracking of glass surfaces shall constitute failure of this test.

4.7.2.6.2 Carrying case. The target, stored in its carrying case, shall be subjected to the test specified in 4.7.2.6.1. Evidence of sand or dust in the carrying case shall constitute failure of this test.

4.7.2.7 Operational. The operational test shall be conducted after each of the following tests:

4.7.2.4.2 Levelment.
4.7.2.6 Sand and dust.

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4.7.2.8.2 Low temperature operation.
4.7.2.8.3 Low temperature storage.
4.7.2.8.4 High temperature operation.
4.8.2.8.5 High temperature storage.
4.7.2.8.6 Solar radiation.
4.7.2.9 Vibration.
4.7.2.10 Shock.

The target set shall be operated as follows to determine its condition:

a. Manipulate all moving parts and inspect for freedom of movement and non-interference with other parts in various operating positions throughout their maximum ranges.
b. Fit or attach all accessories to the target. Check for functionality and check for voltage output of battery box not to exceed 3 volts.
c. Check levels for free movement to determine if the bubble remains within the graduations and if the bubble ends remain visible.
d. Rotate the target on its vertical axis for uniform and steady operation.
e. Check parts, components, accessories for damage and deterioration.

Evidence that the target, components or accessories will not function as specified shall constitute failure of this test.

4.7.2.8 Environmental. Tripods used shall be supplied with the set as specified in 6.2 (see 3.1.1).

4.7.2.8.1 Low and high temperature test chamber. The low and high temperature tests shall be conducted in a test chamber of a size that will house the target mounted on a tripod, and also the carrying case and accessories, and will permit the inspector to walk around the periphery of the tripod legs while conducting the examination and tests specified herein.

4.7.2.8.2 Low temperature operation. The target shall be mounted on a tripod. The target, carrying case, battery case (excluding batteries) and accessories shall be placed in the temperature test chamber. The temperatures in the chamber shall be gradually lowered to minus 25° F (minus 32° C) plus or minus 2° F (1° C) and maintained at that temperature until the target is in thermal equilibrium. Batteries (not subject to this test) shall then be placed in the battery box, and the electrical cable from the illumination assembly shall be connected to the battery box. The target shall then be subjected to the operational test specified in 4.7.2.7. Nonconformance to 4.7.2.7 shall constitute failure of this test.

4.7.2.8.3 Low temperature storage. Upon completion of the low temperature operational test, the target shall be placed in its carrying case. The tripod,

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if part of the specified equipment, shall be folded, and all accessories shall be stowed for transport. Batteries shall be removed from the battery box. The temperature of the test chamber shall then be lowered to minus 28° F (minus 33° C) plus or minus 2° F (1° C) and maintained at this temperature for 6 hours. The temperature shall then be gradually raised to that of the laboratory. The target shall be dried out and subjected to the operational test specified in 4.7.2.7. Nonconformance to 4.7.2.7 shall constitute failure of this test.

4.7.2.8.4 High temperature operation. The target mounted on a tripod, carrying case, battery box (with batteries) and accessories shall be placed in the temperature test chamber. The chamber temperture shall then be gradually raised to plus 110° F (plus 43° C) plus or minus 2° F (1° C) and maintained at this temperature until the target is in thermal equilibrium. The target case and accessories shall then be tested as specified in 4.7.2.7. Nonconformance to 4.7.2.7 shall constitute failure of this test.

4.7.2.8.5 High temperature storage. Upon completion of the high temperature operational test, the target shall be placed in its carrying case. The tripod shall be folded and all accessories shall be stored for transport. Batteries shall be removed from the battery box. The temperature of the test chamber shall then be raised to plus 145° F (63° C) plus or minus 2° F (1° C) and maintained at that temperature for 1 hour. The temperature shall then be gradually lowered to that of the laboratory and the target shall again be subjected to the operational tests specified in 4.7.2.7. Nonconformance to 4.7.2.7 shall constitute failure of this test.

4.7.2.8.6 Solar radiation. The target shall be temperature stabilized at plus 110° F (plus 43° C) plus or minus 2° F (1° C) and a source of directed, radiant energy simulating the spectrum of solar radiation of 355 British thermal units per square foot per hour shall be applied for a period of 4 hours. At the end of this period, the target shall be tested as specified in 4.7.2.7. In addition, the target, case, and accessories shall be examined for peeled, cracked or blistered paint. Presence of any defects in the finish or nonconformance to 4.7.2.7 shall constitute failure of this test.

4.7.2.8.7 Wiring. Temperature test of the wiring shall be performed on the hand lamp and electrical cables. Temperature tests of the wiring may be conducted concurrently with temperature tests of the target, case, and other accessories.

4.7.2.8.7.1 Low temperature wire operation. The electrical and hand lamp cables shall be stored in the battery box when placed in the temperature test chamber. The two cables shall remain in the control chamber during the low temperature operation test of the target. At the end of the operational check of the target

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and without being removed from the chamber the two cables shall be wound around a 3/8 inch mandrel at a rate of 12 turns per minute. The two cables shall then be removed from the chamber and examined for cracks or flaws under a glass having a magnification of 3 diameters. Evidence of damage such as cracking, peeling or separating shall constitute failure of this test.

4.7.2.9 Vibration. The target, secured in its carrying case, shall be rigidly mounted to the vibration table and subjected to two sweeps of sinusoidal cycling vibration in each of three mutually perpendicular axes. Each sweep shall be less than 5 but not more than 55 and back to less than 5 Hertz and traversed in 15 minutes. The vibration amplitude shall be 0.10 inch double amplitude from 5 to 17 Hertz, and 1.5 g from 17 to 55 Hertz. Upon completion of this test, the target shall be mounted on its tripod and tested for operation as specified in 4.7.2.7. Nonconformance to 4.7.2.7 shall constitute failure of this test.

4.7.2.10 Shock. The target in the carrying case shall be flat dropped in an upright position from a height of 6 inches (152.4 mm) onto a level wood floor. The target shall then be examined and given the operational test specified in 4.7.2.7. Nonconformance to 4.7.2.7 shall constitute failure of this test.

4.8 Inspection comparison. The Government may select target sets at any time during the contract production period and subject these target sets to the examination and tests specified in 4.5 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on target sets selected at random from those which have been accepted by the Government and will not include the previously inspected first article and initial production target sets. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.8.1 Inspection failure. Failure of an inspection comparison target set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.8 shall be cause for rejection of the inspection comparison target set and shall be cause for refusal by the Government to continue acceptance of production target sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on target sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all target sets accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.9 Inspection of packaging.

4.9.1 Quality conformance inspection of pack.

4.9.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.9.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.9.1.3 Examination. Samples selected in accordance with 4.9.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

125.	Methods or containers not as specified.	Each incorrect method or container shall be considered one defect.
126.	Consolidation of preserved components not as specified.	
127.	Strapping not as specified.	
128.	Marking illegible, incorrect, incomplete, or missing.	

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Components of the target set shall be preserved in accordance with MIL-S-3534, except that the case containing the tribrach, target, and illuminator shall be considered as the inner container in lieu of the inner container specified in MIL-S-3534 for the tribrach with optical plumb and base plate.

5.1.1.1 Consolidation. The preserved components comprising a complete target set shall be consolidated in a close-fitting box conforming to PPP-B-636, W5C, style optional.

5.1.2 Commercial. Each complete target set shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).

5.2.1 Level A. Target sets, preserved as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

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5.2.2 Level B. Packing shall be as specified for level A except that boxes shall be domestic type and strapping shall be finish A.

5.2.3 Commercial. Each complete target set shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The target set is used as a sighting object with theodolites for second- and third-order missile surveys, traverses, triangulation, and azimuth observations. It is essential that the target set, and the theodolite with which it will be used, be interchangeable from the same tribrach in conformance with standard surveying field practice.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Optional accessories required (see 3.1.1).
c. Make, model, type, of theodolite to be used with the target set (see 3.1.2).
d. When a first article is required for inspection and approval (see 3.2).
e. When initial production inspection is required (see 3.3).
f. When battery box shall not be treated or finished (see 3.9).
g. Color required (see 3.16).
h. When Government-loaned property is required (see 3.19).
i. Degree of preservation and degree of packing required (see 5.1 and 5.2).
j. When finish A strapping is required (see 5.2.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction target set. The preproduction model should consist of one or more units. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.19, when applicable (see 6.2).

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6.5 Sand and dust. The sand and dust to be used in the sand and dust test is commercially known as "140-mesh silica flour". Samples produced by the Fenton Foundry Supply Company, Dayton, OH 45403 and Ottawa Silica Company, Ottawa, IL 61350, or their equivalent are satisfactory.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each target set.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of target sets.

6.8 Recycled material. The use of recycled material is encouraged when practical, as long as it meets the requirements of this specification (see 3.4).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-52823A	Target Set, Surveying

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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